

Core Templates — High-Level Reference

The single most generic, abstract template for each question type — the reusable skeleton you adapt per problem. Worked solutions, quick-reference tables, and the full problem index live in `template.md` / each category file.

Conventions, Skeleton & Core API

Every template file in this folder follows the same two-part structure:

1. **Template algorithm + template code** — the reusable pattern, with specific LeetCode problems used as concrete examples inside the code.
 2. **Quick reference table + question/solution** — a table listing every problem (with a *Variation from Template* column), followed by each problem's full solution where deviations are marked `// ← VARIATION: explanation`.
-

Generic Test Harness

Use this boilerplate to run any single problem locally. Replace `data / solve()` with the problem's input and logic. Work through the six steps in the comment before coding.

```
class Question {  
  
    // — PROBLEM-SOLVING PROCESS —————  
    // 1. INPUT / OUTPUT : types, sizes, ranges; what to return; examples.  
    // 2. HIGH-LEVEL ALGO : pick the pattern (two-ptr / BFS / DP / ...); state the idea in 1 line.  
    // 3. IMPLEMENTATION : data structures, loop/recursion, invariants.  
    // 4. EDGE CASES      : empty, size 0/1, duplicates, negatives, overflow, all-same, not-found.  
    // 5. COMPLEXITY      : time & space; is it within limits?  
    // 6. OPTIMIZATION    : reduce time/space (hashing, two-pointer, rolling array, pruning, early exit  
    // —————  
  
    private int data;  
  
    public Question(int data) {  
        this.data = data;  
    }  
  
    public void solve() {  
        // TODO  
    }  
}  
  
public class Solution {  
  
    public static void main(String[] args) {  
        Question q = new Question(0);  
        q.solve();  
    }  
}
```

Naming Conventions (enforced across all template files)

Context	Convention
Index variables	i , j , k (never left , right , low , mid , high)
Linked list	sentinel , previous , current
Queue	Queue<Integer> queue = new ArrayDeque<>();
BFS over tree	Queue<TreeNode> queue = new ArrayDeque<>();
Grid directions	int[] dr = {1,-1,0,0}, dc = {0,0,1,-1}; (DOWN, UP, RIGHT, LEFT); iterate for (int dir=0; dir<4; dir++) using i+dr[dir] , j+dc[dir]
Character frequency	count[ch - 'a']++;
Digit frequency	count[ch - '0']++;
Char to integer build	num = num * 10 + (ch - '0');

Common Java API Cheat-Sheet

The APIs used over and over across these templates. Grouped by what you reach for.

Initialize the core structures

```
Map<Integer, Integer> map    = new HashMap<>();           // key → value
Map<Integer, List<Integer>> adj = new HashMap<>();       // adjacency / buckets
Set<Integer> set            = new HashSet<>();          // membership
List<Integer> list          = new ArrayList<>();         // dynamic array

Deque<Integer> stack = new ArrayDeque<>();              // STACK: push / pop / peek
Deque<Integer> queue = new ArrayDeque<>();              // QUEUE: offer / poll / peek
// (ArrayDeque beats java.util.Stack / LinkedList)

PriorityQueue<Integer> minHeap = new PriorityQueue<>();  // min-heap (default)
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq        = new PriorityQueue<>((a, b) -> a[1] - b[1]); // by field, min-first

TreeMap<Integer, Integer> tmap = new TreeMap<>();       // sorted keys: floor/ceiling/first/last
TreeSet<Integer> tset          = new TreeSet<>();       // sorted set: floor/ceiling/higher/lower
LinkedHashSet<Integer> lset    = new LinkedHashSet<>(); // insertion-ordered set (LFU buckets)
Random rand                    = new Random();          // rand.nextInt(n) → 0..n-1 (RandomizedSet)

int[]    arr = new int[n];           // defaults to 0
boolean[] seen = new boolean[n];     // defaults to false
int[][] grid = new int[m][n];        // 2D, all 0
int[] dr     = {1,-1,0,0}, dc = {0,0,1,-1}; // DOWN,UP,RIGHT,LEFT; iterate for(dir=0..3) i+dr[dir],
```

Map / Set / List operations

```
// — Map —
map.put(k, v);
map.getOrDefault(k, 0); // safe read with default
map.merge(k, 1, Integer::sum); // ← counting frequency (most common)
map.computeIfAbsent(k, x -> new ArrayList<>()).add(v); // ← build adjacency / buckets
map.putIfAbsent(k, v); // ← write only first occurrence (keep earliest in
map.containsKey(k); map.remove(k);
for (Map.Entry<Integer,Integer> e : map.entrySet()) { e.getKey(); e.getValue(); e.setValue(v); }
map.keySet(); map.values();

// — Set —
set.add(x); set.contains(x); set.remove(x);

// — List —
list.add(x); list.get(i); list.set(i, x);
list.remove(list.size() - 1); // ← pop in backtracking
list.size(); list.isEmpty(); list.addAll(other);
Collections.sort(list); // ascending
list.sort((a, b) -> b - a); // custom / descending
Collections.reverse(list); // ← reverse path after building it backwards
```

Stack / Queue / Heap (all via the same two ops)

```
// Deque as STACK (LIFO) // Deque as QUEUE (FIFO)
stack.push(x); queue.offer(x);
int top = stack.pop(); int head = queue.poll();
int look = stack.peek(); int look = queue.peek();
stack.isEmpty(); queue.isEmpty();

// Deque BOTH ENDS — needed for the MONOTONE DEQUE (sliding-window max/min)
// and for addFirst-style level building (zigzag BFS).
deque.offerLast(x); deque.pollLast(); deque.peekLast(); // back (push/pop/peek tail)
deque.offerFirst(x); deque.pollFirst(); deque.peekFirst(); // front (push/pop/peek head)
deque.addLast(x); deque.addFirst(x); // same as offer*, throw if capacity-bou

// PriorityQueue (heap)
pq.offer(x); int best = pq.poll(); int look = pq.peek(); pq.size();
pq.remove(x); // ← O(n) removal of an arbitrary element (lazy-deletion alternative in #480)
```

Integer / Long limits & overflow-safe idioms

```
Integer.MAX_VALUE; // 2147483647 (2^31 - 1) - sentinel for "min so far"
Integer.MIN_VALUE; // -2147483648 (-2^31) - sentinel for "max so far"
Long.MAX_VALUE; // when sums can overflow int, accumulate in long
int k = i + (j - i) / 2; // ← avoid (i + j) overflow in binary search
Integer.compare(a, b); // ← overflow-safe comparator; use instead of (a - b)
(a, b) -> Integer.compare(a[1], b[1]); // ← safe PriorityQueue/Arrays.sort comparator
Double.compare(a, b); // ← comparator for double[] heaps (probability, median)
```

Char ↔ int ↔ String conversions

```
ch = 'a'; // 'a'..'z' → 0..25 (lowercase bucket index)
ch = '0'; // '0'..'9' → 0..9 (digit value)
(char) ('a' + i); // 0..25 → 'a'..'z'
num = num * 10 + (ch - '0'); // ← build a number digit by digit

Integer.parseInt("123"); // String → int (throws on bad input)
Integer.valueOf(123); // int → Integer (boxed)
String.valueOf(123); // int/char/etc → String
Character.getNumericValue('7'); // char → 7

// char classification
Character.isDigit(ch); Character.isLetter(ch); Character.isLetterOrDigit(ch);
Character.isWhitespace(ch); Character.toLowerCase(ch); Character.toUpperCase(ch);
```

String & StringBuilder

```
s.length(); s.charAt(i); s.substring(i, j); // [i, j)
s.toCharArray(); s.split(" "); s.equals(t); s.compareTo(t);
s.indexOf("ab"); s.isEmpty();
s.startsWith("ab"); s.endsWith("z"); s.contains("x"); // prefix / suffix / substring tests
s.toLowerCase(); s.toUpperCase(); // case-normalize (e.g. case-insensitive compare)
s.trim(); s.replace('a', 'b'); s.repeat(n); // strip ends / swap chars / repeat

StringBuilder builder = new StringBuilder();
builder.append(x); builder.reverse(); builder.toString();
builder.charAt(i); builder.setCharAt(i, c); builder.deleteCharAt(i); builder.length();
String.join(",", list); // list → "a,b,c"
```

Arrays utilities

```
Arrays.sort(arr); // ascending in place
Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // 2D by first field
Arrays.fill(dp, Integer.MAX_VALUE); // seed a dp/dist array
Arrays.equals(window, need); // ← element-wise array compare (anagram check)
Arrays.copyOf(arr, len); Arrays.copyOfRange(arr, i, j);
System.arraycopy(src, srcPos, dst, dstPos, len); // ← fast block copy (merge step)
Arrays.asList(1, 2, 3); // fixed-size List view
int sum = Arrays.stream(arr).sum();
int max = Arrays.stream(arr).max().getAsInt();

// List<Integer> ↔ int[]
int[] a = list.stream().mapToInt(Integer::intValue).toArray();
List<Integer> l = Arrays.stream(a).boxed().collect(Collectors.toList());
```

Math & bit operations

```
Math.max(a, b); Math.min(a, b); Math.abs(x);
Math.pow(a, b); Math.sqrt(x); Math.ceil(x); Math.floor(x);

n & (n - 1); // clear lowest set bit
n & (-n); // isolate lowest set bit
1 << i; // bit i set (use 1L << i for i ≥ 31)
(n >> i) & 1; // read bit i
(n & 1) == 0; // ← even (last bit 0)
(n & 1) == 1; // ← odd (last bit 1)
n >> 1; // ← divide by 2 (drop last bit); n << 1 = multiply by 2
Integer.bitCount(n); // # of set bits
Integer.toBinaryString(n);
```

Sorted-structure navigation (TreeMap / TreeSet)

```
tmap.firstKey(); tmap.lastKey();
tmap.floorKey(x); // largest key ≤ x      tmap.ceilingKey(x); // smallest key ≥ x
tmap.lowerKey(x); // largest key < x      tmap.higherKey(x); // smallest key > x
tset.floor(x); tset.ceiling(x); tset.lower(x); tset.higher(x); tset.first(); tset.last();
```

Optimization idioms

Common speed/space wins used throughout the templates.

```
// — NEGATIVE-SAFE MODULO — replaces the manual ((x % n) + n) % n
Math.floorMod(x, n); // always in [0, n-1] even when x < 0 (prefix-sum % k)

// — int[] INSTEAD OF HashMap for a bounded key space —
int[] count = new int[26]; // 0(1) no-boxing freq vs HashMap<Character,Integer>
count[ch - 'a']++; // also faster to compare: Arrays.equals(a, b)

// — 1D ROLLING ARRAY for DP — drop a dimension when row i only needs row i-1
int[] dp = new int[n + 1]; // 0(n) space instead of 0(m·n)
for (...) for (int j = target; j >= w; j--) dp[j] += dp[j - w]; // 0/1 knapsack: iterate j DOWN

// — FIXED-SIZE HEAP for top-k — 0(n log k) beats sorting 0(n log n)
if (heap.size() > k) heap.poll(); // keep only k best; min-heap for "k largest"

// — LAZY DELETION over pq.remove(obj) — pq.remove is 0(n); skip stale entries instead
if (entry[0] != dist[node]) continue; // Dijkstra: ignore outdated heap entries

// — ArrayDeque over Stack / LinkedList — faster, no synchronization, both ends 0(1)
Deque<Integer> stack = new ArrayDeque<>(); // never use java.util.Stack

// — StringBuilder over String += in a loop — 0(n) vs 0(n²)
StringBuilder builder = new StringBuilder(); // append, then builder.toString() once

// — EARLY EXIT / PRUNING — return the moment the answer is decided
if (found) return result; // e.g. backtracking bounds
```

Template Index

File	Category	Core Pattern
binary_search.md	Binary Search	Closed <code>[i, j]</code> , half-open, minimize/maximize answer space
two_pointers.md	Two Pointers	Opposite ends, same-direction write pointer, Dutch flag
prefix_sum_hashmap.md	Prefix Sum	Count/length with HashMap, tree path sums
sorting.md	Sorting	Merge sort, quick sort, quick select, count-during-merge
sliding_window.md	Sliding Window	Fixed, max-variable, min-variable windows
linked_list.md	Linked List	Delete/filter, reversal, fast/slow, merge, composite
dfs.md	DFS / Backtracking	Tree return-up/pass-down, grid DFS, graph 3-color, backtracking, BST
bfs.md	BFS	Level-order with size snapshot, position indexing
graph.md	Graph	BFS shortest path, topo sort, union find, Dijkstra, bipartite, Prim's MST
dynamic_programming.md	Dynamic Programming	1D/2D/grid/interval DP, knapsack, state machine
stack_queue.md	Stack / Queue / Heap	Monotone stack/deque, two heaps, priority queue
greedy.md	Greedy	Sort-by-end, sort-by-start, sort + heap, non-interval
string.md	String	Frequency counting, hashing, bucket sort, expand-from-center
bit_manipulation.md	Bit Manipulation	XOR cancel, mod-k state machine, bitmask as set, power checks
design.md	Design	HashMap + linked list (LRU), frequency maps (LFU), array + map
math.md	Math	Fast power, base conversion, sieve, digit manipulation

Two Pointers `two_pointers.md`

Two patterns — pick based on whether pointers converge or march together.

Pattern 1 — Opposite Two Pointers

Variables: `i` = left pointer · `j` = right pointer · scan range `[i, j]` closed **Pseudocode:**

```
set i to the left end, j to the right end
while the pointers haven't crossed:
  if the current pair hits the target:
    record it, then step both ends inward
  else if the value is too small:
    move the left end right to grow it
  else (too large):
    move the right end left to shrink it
```

```
// MENTAL MODEL: two ends close in on the answer; each comparison rules out one whole end of the range.
// WHY sorted: moving inward only safely eliminates half the space if the array is sorted
int i = 0, j = n - 1;
while (i < j) {
    if (condition == target) {
        /* record */ i++; j--;
    } else if (tooSmall) {
        i++;
    } else {
        j--;
    }
}
```

Sort first. `i` and `j` converge inward. Used when the array is sorted and you can eliminate half the search space each step.

Sliding Window `sliding_window.md`

Three variants — pick by what the problem asks for.

`i` = left boundary (inclusive), `j` = right boundary (loop variable, inclusive).

Window = `[i, j]`, size = `j - i + 1`.

The Three Templates

Variables: `i` = left edge (inclusive) · `j` = right edge / loop variable (inclusive), window = `[i, j]`, size = `j - i + 1` · `n` = array length · `k` = target window size · `result` = answer (max length, or min length seeded to MAX) **Pseudocode:**

FIXED window of size `k`:

```
i = 0
for j = 0..n-1:
    add nums[j] to window state
    if window size (j - i + 1) == k:
        record the window (it is exactly size k)
        remove nums[i] from state; i = i + 1
```

MAX variable window (longest valid):

```
i = 0; result = 0
for j = 0..n-1:
    add nums[j] to state
    while window violates the constraint:
        remove nums[i] from state; i = i + 1
    result = max(result, j - i + 1)    (window is valid here)
```

MIN variable window (shortest valid):

```
i = 0; result = +infinity
for j = 0..n-1:
    add nums[j] to state
    while window is valid:
        result = min(result, j - i + 1)    (record while still valid)
        remove nums[i] from state; i = i + 1    (then shrink to try smaller)
```

```

// FIXED window of size k – shrink exactly once when full
// a constant-width window slides one step at a time; add the new cell, drop the old one. – WHEN: "sub
int i = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    if (j - i + 1 == k) {
        // 2. record (window is exactly size k)
        // 3. remove nums[i] from state
        i++;
    }
}

// MAX variable window – find LONGEST valid window
// grow greedily; shrink only enough to restore validity, then the window is the best ending here. – W
int i = 0, result = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window exceeds constraint */) { // shrink WHILE window exceeds constraint
        // remove nums[i] from state
        i++;
    }
    result = Math.max(result, j - i + 1); // record after window is valid
}

// MIN variable window – find SHORTEST valid window
// grow until valid, then shrink aggressively while still valid to find the tightest fit. – WHEN: "sho
int i = 0, result = Integer.MAX_VALUE;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window VALID */) { // record then shrink WHILE window still valid
        result = Math.min(result, j - i + 1); // record while valid
        // remove nums[i] from state
        i++; // then try to shrink
    }
}

```

Frequency Map Trick (used by 567 and 76)

Both use a `need[]` array and a `required` counter to avoid scanning the map each step.

Variables: `need[c]` = how many more of char `c` the window still needs (positive = still short, zero/negative = surplus) ·
`required` = total chars still missing; hits 0 when the window covers all of t · `i` = left edge, `j` = right edge

Pseudocode:

Expanding (add char at j):

```

if need[char] was still positive (> 0): required = required - 1    (filled one more)
need[char] = need[char] - 1

```

Shrinking (remove char at i):

```

if need[char] was already 0 or surplus (>= 0): required = required + 1    (now short one)
need[char] = need[char] + 1
i = i + 1

```



```
// Expanding j - add s.charAt(j):
if (need[s.charAt(j)]-- > 0) required--; // was needed → satisfied one more

// Shrinking i - remove s.charAt(i):
if (need[s.charAt(i)]++ >= 0) required++; // was 0 (used up) → now short one
i++;
```

`need[c]` is positive when still needed, zero or negative when excess.
`required` reaches 0 exactly when all characters of `t` are covered.

Prefix Sum + HashMap `prefix_sum_hashmap.md`

Core idea: subarray sum $[l, r] = \text{prefix}[r+1] - \text{prefix}[l]$.

To find a subarray with a target property, rephrase as: "has a prefix with value `X` been seen before?"

Store that `X` in a HashMap for $O(1)$ lookup.

Core Template

Variables: `map` = prefixSum → count or index seen so far · `sum` = running prefix sum · `lookup` = the complement prefix we search for · `i` = current scan position **Pseudocode:**

```
create empty map of prefix → (count or index)
seed map with prefix 0 (count=1 for counting, index=-1 for length)
sum = 0
for each index i:
    add nums[i] to sum
    lookup = transform(sum) // sum-k, sum%k, etc.
    use map.get(lookup) to update result
    store sum (or sum%k) into map with count or index i
```

```
// a subarray is the difference of two prefixes; remember prefixes seen so far and look up the compleme
Map<Long, Integer> map = new HashMap<>();
map.put(0L, ???); // seed: empty prefix (sum=0) seen before index 0
long sum = 0;
// result = 0 (count) or Integer.MAX_VALUE (length sentinel) depending on problem

for (int i = 0; i < nums.length; i++) {
    sum += nums[i];
    long lookup = transform(sum); // what to look up (sum-k, sum%k, etc.)
    // use map.get(lookup) → update result
    map.put(storeKey(sum), storeValue(i)); // what to store (sum or sum%k → count or index)
}
```

What changes per problem:

Variable	Count problems (560, 974)	Length problems (325, 523)
Map value	<code>count</code> (how many times seen)	<code>first index</code> (earliest occurrence)
Seed value	<code>map.put(0L, 1)</code>	<code>map.put(0L, -1)</code>
On lookup hit	<code>count += map.get(lookup)</code>	<code>maxLen = max(i - map.get(lookup))</code>
Overwrite map?	Yes (merge count)	No (keep first index only)

Binary Search

binary_search.md

Binary search appears in two forms: searching an **array index**, or searching an **answer value**.

The Only Template — [i, j), find first k with condition(k)

There is one binary search. Define a monotone `condition(k)` (`false...false` `TRUE...true`); the loop returns the **first k where `condition(k)` is true**. Then look at `i`: if `i == n` no index qualified, otherwise `i` is the boundary — check the value to decide.

Variables: `[i, j)` = half-open search range (answer lives in here) · `k` = midpoint · `condition(k)` = monotone test, `false...false` then `TRUE...true`; answer is the first `k` it holds **Pseudocode:**

```
i = 0, j = length      # half-open [i, j)
while range non-empty (i < j):
    k = midpoint of i and j
    if condition(k) is false: answer is to the right, set i = k+1
    else: k might be the answer, set j = k
```

```
int i = 0, j = nums.length;      // [i, j)
// find first k that meets condition(k)
while (i < j) {
    int k = i + (j - i) / 2;
    if (!condition(k)) {          // condition false → answer is strictly to the right
        i = k + 1;
    } else {                      // condition true → k might be the answer
        j = k;
    }
}
// i in [0, nums.length]; if i == nums.length, no k met condition(k)
```

Pick `condition(k)`:

Want	condition(k)	Answer after loop
lowerBound — first <code>>= target</code>	<code>nums[k] >= target</code>	<code>i</code>
upperBound — first <code>> target</code>	<code>nums[k] > target</code>	<code>i</code>
exact search (#704)	<code>nums[k] >= target</code>	<code>i < n && nums[i] == target ? i : -1</code>
insert position (#35)	<code>nums[k] >= target</code>	<code>i</code>
first / last occurrence (#34)	<code>>=</code> then <code>></code>	<code>lowerBound, upperBound - 1</code>
count of target	—	<code>upperBound - lowerBound</code>
minimize feasible X	<code>feasible(k)</code> over value range	<code>i</code>
maximize feasible X	<code>!feasible(k)</code> (first infeasible)	<code>i - 1</code>

The rule: the answer is always `i`. Guard `i == nums.length` (nothing met the condition) before reading `nums[i]`, then confirm the value.

How to Choose

Signal	condition(k)
Sorted array, find exact value	<code>arr[k] >= target</code> , then check <code>arr[i] == target</code>
Insert position / first occurrence	<code>arr[k] >= target</code> → return <code>i</code> (= <code>lowerBound</code>)
Last occurrence / count	<code>arr[k] > target</code> (= <code>upperBound</code>); last = <code>upperBound - 1</code> , count = <code>upper - lower</code>
Rotated / peak slope search	<code>condition(k)</code> compares <code>arr[k]</code> to a neighbor or endpoint
"minimum X such that feasible(X)"	<code>condition(k) = feasible(k)</code> over value range → return <code>i</code>
"maximum X such that feasible(X)"	<code>condition(k) = !feasible(k)</code> over <code>[lo, hi+1)</code> → return <code>i - 1</code>

The one rule: define `condition(k)` so it is monotone `false...false TRUE...true` , then the answer is the first `k` that flips it true (`j = k` on true, `i = k + 1` on false). For MAXIMIZE, make the condition "first infeasible" and subtract 1 — same loop, no ceiling mid.

Sorting `sorting.md`

All recursive sorting uses **half-open intervals** `[start, end)` throughout.

Inside merge and partition: `i` , `j` , `k` as scan/write pointers (consistent with binary search convention).

Template 1 — Merge Sort

Variables: `[start, end)` half-open range being sorted · `mid` = split point · `temp` = scratch buffer · `i` = left scan pointer `[start,mid)` · `j` = right scan pointer `[mid,end)` · `k` = write pointer into temp **Pseudocode:**

```
mergeSort(nums, start, end):
    if range has 0 or 1 elements: return          // already sorted
    mid = midpoint of [start, end)
    mergeSort left half [start, mid)
    mergeSort right half [mid, end)
    merge the two sorted halves

merge:
    i = start, j = mid, k = start
    while either half has elements left:
        pick smaller front of the two halves (left wins ties → stable)
        write it to temp[k], advance that half's pointer and k
    copy temp back into nums[start, end)
```

```
// split in half until trivially sorted, then merge two sorted halves into one. - WHEN: need stable so
// Entry point
public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new int[nums.length]);
}

// [start, end) half-open
private void mergeSort(int[] nums, int start, int end, int[] temp) {
    if (end - start <= 1) return; // base case: end - start <= 1 → 0 or 1 elements, already sort
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp); // sort [start, mid)
    mergeSort(nums, mid, end, temp); // sort [mid, end)
    merge(nums, start, mid, end, temp);
}

// i scans left [start,mid), j scans right [mid,end), k writes to temp
private void merge(int[] nums, int start, int mid, int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j >= end || (i < mid && nums[i] <= nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}
```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

Variables: [start, end) half-open range · pivot = chosen partition value · i = boundary of <pivot region · k = scan/cursor pointer · j = boundary of >pivot region (from the right) **Pseudocode:**

```
quickSort(nums, start, end):
    if range has 0 or 1 elements: return
    pivot = middle element's value
    i = start, k = start, j = end-1 // [start,i)<p | [i,k]==p | [k,j] unknown | (j,end)>p
    while k <= j:
        if nums[k] < pivot: swap into < region, advance k and i
        elif nums[k] == pivot: advance k
        else: swap to > region, shrink j (don't advance k)
    recurse on < region [start, i)
    recurse on > region [k, end) // ==pivot middle is final
```

```

// pick a pivot, partition into <pivot | ==pivot | >pivot, recurse on the two ends only. - WHEN: in-pl
// Entry point - no extra array needed (in-place)
public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}

// [start, end) half-open
private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    // Invariant: [start,i) < pivot | [i,k) == pivot | [k,j) unknown | (j,end) > pivot
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    // After: [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
    quickSort(nums, start, i); // recurse on < region
    quickSort(nums, k, end); // recurse on > region
}

private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}

```

Template 3 — Quick Select (partial sort — k-th element)

Extends quick sort: after partition, only recurse into the side that contains `target` index.

Variables: `[start, end)` half-open range · `target` = 0-indexed position we want settled · `pivot` = partition value · `i` = boundary of `<pivot` region · `k` = scan/cursor pointer · `j` = boundary of `>pivot` region **Pseudocode:**

```

quickSelect(nums, start, end, target):
    pivot = middle element's value
    partition into [start,i)<p | [i,k)==p | [k,end)>p // same 3-way as quick sort
    if target < i:        recurse into left [start, i)
    elif target < k:      target sits in ==pivot region → return pivot
    else:                recurse into right [k, end)

```

```
// same partition as quick sort, but recurse into only the one side holding target → O(n) avg. – WHEN:
// Returns value at 0-indexed position target after sorting
private int quickSelect(int[] nums, int start, int end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    // [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target < k) {
        return pivot; // target lands in == region
    } else {
        return quickSelect(nums, k, end, target);
    }
}
```

Template 4 – Counting Sort

$O(n + \text{range})$. Use when values are bounded integers.

Variables: `min / max` = value range bounds · `buckets[v-min]` = count of value `v` · `i` = write pointer back into `arr` · `v` = current value index into buckets **Pseudocode:**

```
if array empty: return
find min and max values
buckets = array sized (max-min+1), all zero
for each x in arr: increment buckets[x-min] // tally occurrences
i = 0
for v from 0 to last bucket:
    while bucket v still has count: write (v+min) into arr[i], advance i
```

```
// tally how many times each value occurs, then emit values in order – no comparisons. – WHEN: integer
public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}
```

Linked List `linked_list.md`

Four patterns cover almost every linked list problem.

Consistent naming throughout: `sentinel`, `previous`, `current`, `next`, `slow`, `fast`.

When to Use — Signal → Pattern

Signal in the prompt	Pattern
"remove / delete / skip" nodes by value or duplicate	Delete / Filter (sentinel + previous)
"reverse" the list, a sub-range, or in k-groups	Reversal
"middle node", "cycle", "nth node from the end"	Fast / Slow
"merge two sorted lists" (or k lists)	Merge (sentinel + heap for k)
chained steps ("reorder", "palindrome", "sort")	Composite (combine the above)

All Four Templates

Variables: sentinel = dummy node before head · previous = last kept node · current = node under inspection · next = saved successor · slow / fast = 1×/2× walkers · a / b = the two input lists **Pseudocode:**

```
make sentinel pointing at head; previous = sentinel, current = head
while current exists:
    if current should be deleted: rewire previous.next to skip current
    else: advance previous to current
    advance current
return sentinel.next

previous = null, current = head
while current exists:
    save next = current.next
    point current backward at previous
    advance previous to current, current to next
return previous as new head

slow = head, fast = head
while fast and fast.next exist:
    advance slow one step, fast two steps

make sentinel; current = sentinel
while both a and b exist:
    splice the smaller head onto current; advance that list
    advance current
attach whichever list remains
return sentinel.next
```

```

// 1. DELETE / FILTER – sentinel guards head removal; previous tracks last kept node
// previous always points at the last node you decided to keep, so re-wiring it skips anything unwanted
ListNode sentinel = new ListNode(0);
sentinel.next = head;
ListNode previous = sentinel, current = head;
while (current != null) {
    if (shouldDelete(current)) {
        previous.next = current.next;    // skip node
    } else {
        previous = current;              // keep node, advance previous
    }
    current = current.next;
}
return sentinel.next;

// 2. REVERSAL – re-wire one node at a time
// flip each arrow to point backward; previous is the growing reversed list trailing behind current. –
ListNode previous = null, current = head;
while (current != null) {
    ListNode next = current.next;    // save before overwrite
    current.next = previous;         // reverse pointer
    previous = current;              // advance
    current = next;
}
return previous; // new head

// 3. FAST / SLOW – two pointers at different speeds
// fast covers twice the distance, so when it hits the end slow is exactly halfway; in a loop they must
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
}
// slow is at middle (or cycle detection: slow == fast → cycle found)

// 4. MERGE – sentinel collects nodes from two lists in order
// repeatedly splice the smaller head onto a growing result tail, like a zipper. – WHEN: "merge two (o
ListNode sentinel = new ListNode(0);
ListNode current = sentinel;
while (a != null && b != null) {
    if (a.val <= b.val) {
        current.next = a;
        a = a.next;
    } else {
        current.next = b;
        b = b.next;
    }
    current = current.next;
}
current.next = (a != null) ? a : b;
return sentinel.next;

```

`previous` stays on the last **kept** node. When deleting, re-wire `previous.next` to skip `current`. When keeping, advance `previous` to `current`. Always advance `current`.

String `string.md`

Core idioms first, then problems grouped by pattern.

For sliding window string problems (#3, #76, #567, #424), see `sliding_window.md`.

Core Idioms

Variables: `count` = frequency vector (index = char/digit bucket) · `ch - 'a'` = 0–25 bucket index · `ch - '0'` = 0–9 digit value · `num` = number built so far · `builder` = encoded/result string · `buckets` = lists indexed by frequency · `expand(i, j)` = palindrome length from a center **Pseudocode:**

```
idiom 1 – char count for letters:
    make count of size 26
    for each char: count[ch-'a'] += 1
idiom 2 – char count for digits:
    make count of size 10
    for each char: count[ch-'0'] += 1
idiom 3 – char to integer:
    num = 0
    for each char left to right: num = num*10 + (char - '0')
idiom 4 – anagram hash key (frequency-based):
    count the 26 letters of s
    build a string from each letter label followed by its count
    return it (same key for all anagrams)
    alternative: sort the chars and use the sorted string as key
idiom 5 – bucket sort by frequency:
    count all ASCII chars
    make buckets indexed by frequency
    for each char with count>0: add it to buckets[its count]
    walk buckets from highest frequency down, append each char that many times
idiom 6 – expand from center:
    while i in bounds and j in bounds and chars at i and j match: move i left, j right
    return j - i - 1 (length of palindrome found)
```

```

// a lowercase string is a length-26 frequency vector; most string problems are vector ops on it. - WH
// 1. CHAR COUNT - lowercase letters
int[] count = new int[26];
// WHY ch - 'a': ASCII 'a'=97, so 'a'→0, 'b'→1, ..., 'z'→25 - a clean 0-25 bucket index into count[].
for (char ch : s.toCharArray()) count[ch - 'a']++; // ← ch - 'a' maps a→0, b→1, ..., z→25

// 2. CHAR COUNT - digits 0-9
int[] count = new int[10];
for (char ch : s.toCharArray()) count[ch - '0']++; // ← ch - '0' maps '0'→0, '9'→9

// 3. CHAR TO INTEGER - build number digit by digit
int num = 0;
for (int i = 0; i < s.length(); i++)
    num = num * 10 + (s.charAt(i) - '0'); // ← shift left × 10, add new digit

// 4. ANAGRAM HASH KEY - frequency-based (bucket sort style) O(k) vs sort O(k log k)
// WHY it works: anagrams share identical letter frequencies, so encoding the frequency vector
// ITSELF gives a unique key for the whole group - O(k) to build vs O(k log k) to sort the chars.
String hashCode(String s) {
    int[] count = new int[26];
    for (char ch : s.toCharArray()) count[ch - 'a']++;
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < 26; i++) {
        builder.append((char)('a' + i)); // ← letter as label
        builder.append(count[i]); // ← its count
    }
    return builder.toString(); // e.g., "a2b0c1...z0"
}
// Alternative: sort the characters (simpler, slightly slower)
char[] chars = s.toCharArray();
Arrays.sort(chars);
String key = new String(chars); // anagrams → same sorted key

// 5. BUCKET SORT - sort by frequency without comparison sort
int[] count = new int[128]; // ASCII
for (char ch : s.toCharArray()) count[ch]++;
List<Character>[] buckets = new List[s.length() + 1]; // index = frequency
for (int i = 0; i < 128; i++) {
    if (count[i] > 0) {
        int freq = count[i];
        if (buckets[freq] == null) buckets[freq] = new ArrayList<>();
        buckets[freq].add((char) i);
    }
}
StringBuilder builder = new StringBuilder();
for (int freq = buckets.length - 1; freq > 0; freq--) { // ← descending frequency
    if (buckets[freq] == null) continue;
    for (char ch : buckets[freq])
        for (int i = 0; i < freq; i++) builder.append(ch);
}

// 6. EXPAND FROM CENTER - palindrome check in O(1) space
int expand(String s, int i, int j) {
    while (i >= 0 && j < s.length() && s.charAt(i) == s.charAt(j)) { i--; j++; }
    return j - i - 1; // ← length of palindrome found
}

```

Stack / Queue / Heap stack_queue.md

Four data structures — each with a canonical template and representative problems.

All Templates

Variables: `stack` = LIFO ArrayDeque (holds indices for monotone variants) · `queue` = monotone deque (front = window answer) · `minPQ` / `maxPQ` / `pq` = priority queue (heap) · `maxHeap` = lower half, `minHeap` = upper half for median

Pseudocode:

STACK: push/pop/peek all at the front (LIFO)

MONO-DECREASING (next greater): while current > stack top, pop and record current as its answer; push inc

MONO-INCREASING (next smaller / span): while current < stack top, pop and compute width to new top bounda

MONO-DEQUE (window max): drop back while smaller than current, push; drop front if out of window; front =

PRIORITY QUEUE: offer/poll/peek the min (or max with reverse comparator)

TWO HEAPS: push to maxHeap then shift its top to minHeap; rebalance so maxHeap >= minHeap; median from th

```

// 1. STACK (LIFO) – use ArrayDeque, not Stack class
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);    // add to top
stack.pop();      // remove from top
stack.peek();     // view top, no remove

// 2A. MONOTONE DECREASING STACK – next GREATER element
// keep only candidates still waiting for a bigger neighbor; current resolves them all. – WHEN: "next/"
// Invariant: stack values decrease from bottom to top
// Pop when current is GREATER than top → top found its next greater
Deque<Integer> stack = new ArrayDeque<>(); // stores indices
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] < nums[i])
        result[stack.pop()] = nums[i];    // nums[i] is next greater for popped index
    stack.push(i);
}
// Remaining in stack: no next greater element exists

// 2B. MONOTONE INCREASING STACK – next SMALLER element / span width
// each popped bar's reach extends until something shorter blocks it on both sides. – WHEN: "largest r
// Invariant: stack values increase from bottom to top
// Pop when current is SMALLER than top → use popped index to compute width/span
Deque<Integer> stack = new ArrayDeque<>(); // stores indices
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] > nums[i]) {
        int height = nums[stack.pop()];
        int width = stack.isEmpty() ? i : i - stack.peek() - 1; // span between boundaries
        // process height * width
    }
    stack.push(i);
}

// 3. MONOTONE DEQUE – sliding window max/min
// the front is the current window's answer; anything a newer-and-bigger value beats is dead weight. –
// Deque stores INDICES; values at those indices are decreasing front-to-back (for max)
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty() && nums[queue.peekLast()] <= nums[i])
        queue.pollLast();    // remove smaller elements – they can never be window max
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1) queue.pollFirst(); // evict out-of-window elements
    if (i >= k - 1) result[i - k + 1] = nums[queue.peekFirst()];
}

// 4. PRIORITY QUEUE
PriorityQueue<Integer> minPQ = new PriorityQueue<>(); // min-heap (default)
PriorityQueue<Integer> maxPQ = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // custom: sort by index 1
pq.offer(x); // add
pq.poll(); // remove and return min/max
pq.peek(); // view min/max without removing

// 5. TWO HEAPS – maintain median in O(log n)
// split the data at the median; the median is always sitting on the two heaps' tops. – WHEN: "running
// maxHeap = lower half (smaller numbers), minHeap = upper half (larger numbers)
// Invariant: maxHeap.size() == minHeap.size() or maxHeap.size() == minHeap.size() + 1
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> minHeap = new PriorityQueue<>();
void addNum(int num) {
    maxHeap.offer(num);
    minHeap.offer(maxHeap.poll()); // push one to minHeap (possibly num itself)
    if (maxHeap.size() < minHeap.size())
        maxHeap.offer(minHeap.poll()); // rebalance: maxHeap always ≥ minHeap in size
}

```

```

}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}

```

BFS (Tree) bfs.md

Core structure: `Queue<TreeNode> queue = new ArrayDeque<>()`.

One template: snapshot the level size, then drain exactly that many nodes per level.

The Template — Level-Aware BFS

Variables: `queue` = nodes waiting to be processed · `size` = level snapshot (count of nodes in the current level) · `level` = current depth counter · `node` = node polled this step · `i` = index within the level **Pseudocode:**

```

make an empty queue
put root in the queue
level = 0
while the queue is not empty:
    size = how many nodes are in the queue right now (one whole level)
    level = level + 1
    repeat size times (index i = 0..size-1):
        node = take the front of the queue
        process node (level known; i==0 -> first, i==size-1 -> last)
        if node has a left child, add it to the queue
        if node has a right child, add it to the queue

```

```

// LEVEL-AWARE PROCESSING — snapshot size to make per-level decisions
// the queue holds exactly one full level; freeze its size, then drain just that many. — WHEN: "level
Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();          // ← snapshot: all nodes currently in queue = this level
    level++;
    for (int i = 0; i < size; i++) {
        TreeNode node = queue.poll();
        // process node (know: level, i==0 → first, i==size-1 → last)
        if (node.left != null) queue.offer(node.left);
        if (node.right != null) queue.offer(node.right);
    }
}

```

Always snapshot `size` first — it's how you know where each level ends. (If you truly don't need levels, just drop the `size` loop and poll node-by-node.)

When to Use — Signal → Pattern

Problem signal phrase	Pattern / group
"level order", "values per level", "each row"	Group 1 — collect entire level
"average / max / sum of each level"	Group 2 — aggregate per level
"right side view" (last in level), "bottom-left" (first in level)	Group 2 — record boundary node (<code>i==size-1</code> / <code>i==0</code>)
"minimum depth", "first level that satisfies X"	Group 3 — early return on condition
"connect next pointers at the same level"	Group 4 — wire nodes within a level
"same depth, different parent" (cousins), per-node metadata	Group 5 — track per-node metadata
"maximum width of a level" (counting null gaps)	Group 6 — position indexing

The One Rule — Snapshot Level Size First

Always snapshot the level size BEFORE the inner for loop:

```
int size = queue.size();
for (int i = 0; i < size; i++) { ... }
```

Without the snapshot, newly added children mix with the current level and `i == size-1` / `i == 0` checks become meaningless.

DFS / Backtracking `dfs.md`

DFS appears in five forms. The pattern determines naming, return type, and where the answer is built.

When to Use — Signal → Pattern

Problem signal phrase	Pattern / template
"height / depth / sum OF the subtree", combine results upward	#1 Process children first, combine at node (post-order)
"root-to-leaf path", "number built along the path"	#2 Process node first, pass state down (pre-order)
"longest/best path that may bend through a node" (spans both arms)	#3 Tree global answer
"count islands / largest region / flood fill" on a grid	#4 Grid DFS
"can finish all", "detect cycle", "valid course order" (directed)	#5 Graph DFS 3-color
"all subsets / permutations / combinations", "find every way"	#6 Backtracking
"valid BST", "kth smallest", "closest value", "successor"	BST templates (pass bounds / inorder / binary search)

All Templates

Variables: `node` = current tree node · `accumulated / current` = state carried down a path · `answer / global` = best cross-node result · `grid[r][c]` = cell at row `r`, col `c` · `state[node]` = 0 unseen / 1 in-stack / 2 done · `start` = first eligible index · `current` = in-progress candidate · `nums[i]` = element being chosen **Pseudocode:**

```
if node null: return BASE; left=dfs(left); right=dfs(right); return combine(left,right,node.val)
if node null: return; accumulated+=node.val; if leaf: record; dfs(left,accumulated); dfs(right,accumulated)
if node null: return 0; left=max(0,dfs(left)); right=max(0,dfs(right)); answer=max(answer,left+right+node.val)
if out of bounds or not TARGET: return; mark VISITED; recurse into 4 neighbors
if state==1: cycle; if state==2: safe; mark in-stack; recurse neighbors (cycle? return); mark done
if base: record copy; for i from start: skip dups; choose nums[i]; recurse(next); undo
```

```

// 1. PROCESS CHILDREN FIRST, COMBINE AT NODE (post-order)
// each node trusts its children to return a finished answer, then merges them. - WHEN: "what is the h
private int dfs(TreeNode node) {
    if (node == null) return BASE;          // 0 for depth, true for balanced
    int left = dfs(node.left);
    int right = dfs(node.right);
    // combine left, right, node.val → answer propagates up
    return ...;
}

// 2. PROCESS NODE FIRST, PASS STATE DOWN (pre-order)
// carry what you've seen so far down the path; the leaf has the full picture. - WHEN: "root-to-leaf p
private void dfs(TreeNode node, int accumulated) {
    if (node == null) return;
    accumulated += node.val;                // update state on way down
    if (isLeaf(node)) { /* record */ return; }
    dfs(node.left, accumulated);
    dfs(node.right, accumulated);
    // backtrack: no undo needed for primitive; undo List.add if using collection
}

// 3. TREE - GLOBAL ANSWER (return up the best single arm; update global across both arms)
// the answer may bend through a node using both arms, but only one arm can extend upward. - WHEN: use
int answer = Integer.MIN_VALUE;
private int dfs(TreeNode node) {
    if (node == null) return 0;
    int left = Math.max(0, dfs(node.left)); // clamp negative
    int right = Math.max(0, dfs(node.right));
    answer = Math.max(answer, left + right + node.val); // cross-node path
    return Math.max(left, right) + node.val;          // single arm up
}

// 4. GRID DFS - mark visited by mutating grid; 4-directional flood fill
// stand on a cell, flood into all 4 neighbors, sinking each as you visit it. - WHEN: "connected regio
private void dfs(int[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >= grid[0].length) return;
    if (grid[r][c] != TARGET) return;
    grid[r][c] = VISITED;
    dfs(grid, r+1, c); dfs(grid, r-1, c);
    dfs(grid, r, c+1); dfs(grid, r, c-1);
}

// 5. GRAPH DFS - 3-color visited: 0=unseen 1=in-stack 2=done
// if you re-enter a node still on the current stack, you've looped back → cycle. - WHEN: "can finish
int[] state;
private boolean dfs(int node) {
    if (state[node] == 1) return true;    // back edge → cycle
    if (state[node] == 2) return false;   // already safe
    state[node] = 1;
    for (int neighbor : graph.get(node))
        if (dfs(neighbor)) return true;
    state[node] = 2;
    return false;
}

// 6. BACKTRACKING - choose / recurse / undo
// build a candidate one element at a time; undo each choice to explore the next branch. - WHEN: "all
private void backtrack(int start, List<Integer> current) {
    if (baseCase) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition(i)) continue; // prune duplicates
        current.add(nums[i]);           // choose
        backtrack(i+1, current);         // next = i (reuse) or i+1 (no reuse)
    }
}

```



```

    current.remove(current.size()-1); // undo
}
}

```

Graph graph.md

Queue<Integer> queue = new ArrayDeque<>() throughout.

Six algorithms — each with a canonical template and representative problems.

Complexity Legend

BFS / DFS	$O(V + E)$	visit each vertex and edge once
Topo sort (Kahn)	$O(V + E)$	each node enqueued once, each edge relaxed once
Union-Find	$\sim O(n \cdot \alpha(n)) \approx O(n)$	α = inverse Ackermann, effectively constant
Dijkstra	$O((V + E) \log V)$	non-negative weights ONLY

Graph Representation

```

// Adjacency list (most common)
List<List<Integer>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges) {
    graph.get(edge[0]).add(edge[1]);
    graph.get(edge[1]).add(edge[0]); // omit for directed
}

// Weighted adjacency list: int[] = {neighbor, weight}
List<List<int[]>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges)
    graph.get(edge[0]).add(new int[]{edge[1], edge[2]});

// Grid 4-directional neighbors
int[] dr = {1, -1, 0, 0};
int[] dc = {0, 0, 1, -1};

```

Core Templates

Variables: queue = BFS frontier · visited[] = seen flags · level = current distance ring · inDegree[] = remaining prerequisites per node · order = topo result · parent[] / rank[] = union-find forest · dist[] = shortest distance · pq = min-heap on cost · color[] = 2-coloring (-1/0/1) **Pseudocode:**

BFS: mark start visited, enqueue; while queue, snapshot level size, pop each, push unvisited neighbors, increment level
 TOPO: count inDegree per edge, enqueue all zero-inDegree, pop into order and decrement neighbors, enqueue zero-inDegree
 UNION-FIND: init parent[i]=i; find follows parent to root with path compression; union links roots by rank
 DIJKSTRA: dist[src]=0, push src; pop closest, skip if stale, relax each neighbor and push improved
 BIPARTITE: color each component start 0, BFS painting neighbors opposite, return false on same-color class

```

// 1. BFS – shortest path / level order (track distance by level-size snapshot)
// explore in rings of equal distance; snapshot the level size so each ring = one step. – WHEN: "fewes
Queue<Integer> queue = new ArrayDeque<>();
boolean[] visited = new boolean[n];
queue.offer(start);
visited[start] = true;
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();                // ← snapshot: all nodes at this distance
    for (int i = 0; i < size; i++) {
        int node = queue.poll();
        // process node (distance == level)
        for (int neighbor : graph.get(node)) {
            if (!visited[neighbor]) {
                visited[neighbor] = true;
                queue.offer(neighbor);
            }
        }
    }
    level++;
}

// 2. TOPOLOGICAL SORT – Kahn's BFS
// peel off nodes with no remaining prerequisites; if any get stuck, a cycle exists. – WHEN: "valid or
int[] inDegree = new int[n];
for (int[] edge : edges) inDegree[edge[1]]++;
Queue<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (inDegree[i] == 0) queue.offer(i);
List<Integer> order = new ArrayList<>();
while (!queue.isEmpty()) {
    int node = queue.poll();
    order.add(node);
    for (int neighbor : graph.get(node))
        if (--inDegree[neighbor] == 0) queue.offer(neighbor);
}
// order.size() == n → no cycle

// 3. UNION FIND
// each set points to one representative; merge sets by linking roots, query by finding roots. – WHEN:
int[] parent, rank;
void init(int n) {
    parent = new int[n]; rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}
int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]); // path compression
    return parent[x];
}
boolean union(int x, int y) {
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py; py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}

// 4. DIJKSTRA – shortest path (non-negative weights)
// greedily settle the closest unfinished node; non-negative weights guarantee it's final. – WHEN: "sh
int[] dist = new int[n];
Arrays.fill(dist, Integer.MAX_VALUE);
dist[src] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // min-heap on cost

```

```

pq.offer(new int[]{src, 0});
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], d = current[1];
    if (d > dist[node]) continue; // stale entry
    for (int[] nb : graph.get(node)) {
        int next = nb[0], w = nb[1];
        if (dist[node] + w < dist[next]) {
            dist[next] = dist[node] + w;
            pq.offer(new int[]{next, dist[next]});
        }
    }
}

// 5. BIPARTITE CHECK – BFS 2-coloring
// paint neighbors the opposite color; a neighbor that's already your color means an odd cycle. – WHEN
int[] color = new int[n];
Arrays.fill(color, -1);
Queue<Integer> queue = new ArrayDeque<>();
for (int start = 0; start < n; start++) {
    if (color[start] != -1) continue;
    color[start] = 0;
    queue.offer(start);
    while (!queue.isEmpty()) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (color[neighbor] == -1) { color[neighbor] = 1 - color[node]; queue.offer(neighbor); }
            else if (color[neighbor] == color[node]) return false; // conflict
        }
    }
}
return true;

```

Greedy greedy.md

Two interval templates plus non-interval greedy patterns.

Two Interval Templates

MERGE DIRECTION RULE (for merge-style problems like #56, where both keys work):

Sort by START → traverse FORWARD ($i = 0 \rightarrow n-1$), extend the END of last kept

Sort by END → traverse BACKWARD ($i = n-1 \rightarrow 0$), extend the START of last kept

These two are exact mirror images. See #56 for both written out side by side.

(NOTE: this mirror only applies to merging. Template 1 below sorts by end but still sweeps FORWARD – there the goal is counting, not merging.)

TEMPLATE 1 – Sort by END time, sweep forward, track end of last kept interval

Use when: maximizing count of non-overlapping intervals

Greedy insight: earliest-finishing interval leaves most room for future ones

TEMPLATE 2 – Sort by START time, sweep forward, merge when overlapping

Use when: merging/counting overlapping intervals

Greedy insight: processing in start order → each interval only needs to compare with the last merged result

Variables: `intervals` = list of `[start, end]` · `lastEnd` = end of last kept interval · `result` = merged output · `pq` = min-heap of active end times **Pseudocode:**

TEMPLATE 1 (sort by end, count non-overlapping):

```
sort intervals by END time
lastEnd = -infinity
for each interval:
    if interval.start >= lastEnd:    # no overlap
        lastEnd = interval.end
        keep it (count++ or add)
    else:
        skip it (overlap)
```

TEMPLATE 2 (sort by start, merge):

```
sort intervals by START time
for each interval:
    if result empty OR last kept end < interval.start:
        add interval as new
    else:
        extend last kept end = max(last end, interval.end)
```

TEMPLATE 3 (sort by start + min-heap of ends):

```
sort intervals by START time
for each interval:
    if heap nonempty AND earliest end <= interval.start:
        pop heap                # reuse freed resource
    push interval.end           # occupy a resource
heap size = resources needed
```

```

// sort to expose a local greedy choice, then sweep once making the locally-best pick. - WHEN: interval
// TEMPLATE 1: Sort by end, compare start
Arrays.sort(intervals, (a, b) -> a[1] - b[1]); // sort by END time
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) { // start >= lastEnd → no overlap → keep
        lastEnd = interval[1];
        // count++ or add to result
    } else {
        // overlap → skip (or count as removed)
    }
}

// TEMPLATE 2: Sort by start, compare end
Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // sort by START time
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() || result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval); // no overlap → new interval
    } else {
        result.get(result.size()-1)[1] = Math.max(result.get(result.size()-1)[1], interval[1]); // overlap → extend end
    }
}

// TEMPLATE 3: Sort by start + min-heap of end times (multi-resource)
// USE WHEN counting simultaneous/overlapping resources - "minimum rooms", "max concurrent".
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>(); // min-heap of active end times
for (int[] interval : intervals) {
    if (!pq.isEmpty() && pq.peek() <= interval[0]) {
        pq.poll(); // reuse: earliest-ending resource is free
    }
    pq.offer(interval[1]); // occupy a resource until interval[1]
}
// pq.size() = resources needed

```

Dynamic Programming `dynamic_programming.md`

Six template families. Every problem maps to one loop skeleton and one dp-state definition.

All Six Templates

Variables: `dp[i]` = the answer for the subproblem at index/state `i` (meaning varies per family:

ways/cost/length/feasibility ending at or reaching `i`) · `n` = problem size · `target` / `j` = knapsack capacity dimension.

Pseudocode:

```

LINEAR 1D:    dp[i] = fixed recipe over dp[i-1], dp[i-2] (Kadane: best ending at i = max(start fresh, extend))
LOOK-BACK:    for each i, scan all earlier j and extend the best valid dp[j]
KNAPSACK:     collapse item dimension to 1D; descending j = each item once, ascending j = reusable
2D SEQUENCE:  match consumes both strings (diagonal), mismatch drops one side
INTERVAL:     solve short ranges first, combine via a split point inside the range
GRID:         each cell accumulates from cells it could arrive from (up / left)
STATE MACHINE: name a few states, update each from yesterday's states every step

```

```
// 1. LINEAR 1D – each cell depends on a fixed number of previous cells
// the answer at i is a fixed recipe over the last one or two answers. – WHEN: "ways/cost to reach ste
int[] dp = new int[n + 1];
dp[0] = base;
for (int i = 1; i <= n; i++)
    dp[i] = f(dp[i-1], dp[i-2], ...);
// KADANE (max subarray, #53) = Linear-1D where dp[i] = best sum ENDING at i:
// dp[i] = max(nums[i], dp[i-1] + nums[i]) // start fresh vs extend (drop a negative prefix)
// answer = max over all dp[i]; collapse dp[] to one rolling int → O(1) space.
// variants: #152 product (track max AND min, a negative swaps them); #918 circular.

// 2A. LOOK-BACK 1D – dp[i] depends on all j < i
// to finish position i, scan every earlier position j and extend the best valid one. – WHEN: "longest
int[] dp = new int[n];
for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        dp[i] = f(dp[i], dp[j]); // e.g., LIS, word break
    }
}

// 2B/2C. KNAPSACK – collapse the item dimension to 1D; loop direction picks reuse.
// 0/1 (each item once): j DESCENDING → reads previous row.
// unbounded (reusable): j ASCENDING → reads current row.
// Mnemonic: DESCENDING = Distinct, ASCENDING = Again. Full code + permutation
// variant + "why" in Part 2 – Knapsack DP.

// 3. 2D SEQUENCE – match consumes both, mismatch drops one side. WHEN: LCS / edit
// distance / subsequence count over two strings. Full code → Part 3.
// 4. INTERVAL DP – short ranges first (i right-to-left, j from i+1; dp[i+1][*] ready).
// WHEN: palindrome / merge-burst / split-point in a range. Full code → Part 4.
// 5. GRID DP – each cell accumulates from cells it could arrive from (up/left).
// WHEN: grid paths / min-cost / largest square. Full code → Part 5.
// 6. STATE MACHINE – named states updated from yesterday's states each step.
// WHEN: buy/sell/hold, limited transactions, cooldown/fee. Full code → Part 6.
```

Knapsack Decision Tree

```
Want count or min/max?
├─ Count (dp[j] += dp[j - num])
│   ├── 0/1 (each item once):    j descending → #416, #494
│   └─ Unbounded (reuse):        j ascending  → #518
│       └─ Ordered (permut):      target outer, nums inner → #377
└─ Min/Max (dp[j] = min/max(...))
    ├── 0/1 minimize:             j descending → #474 (maximize)
    └─ Unbounded minimize:        j ascending  → #322
```

Part 2 — Knapsack DP

Variables: `dp[j]` = number of ways to reach sum `j` (count flavor; `dp[0]=1` is the empty selection); `i` = item index, `j` = current capacity/target, `num` = an item's weight/value. Loop *direction* decides reuse: descending `j` = item used at most once, ascending `j` = item reusable; target-outer = orderings counted. **Pseudocode:**

0/1 KNAPSACK (each item once):

```
dp[0] = 1
for each item i:
    for j from target DOWN TO nums[i]:    // DESCENDING reads OLD dp (item i not yet used)
        dp[j] += dp[j - nums[i]]
```

UNBOUNDED KNAPSACK (item reusable):

```
dp[0] = 1
for each item i:
    for j from nums[i] UP TO target:    // ASCENDING reads NEW dp (item i already used this pass)
        dp[j] += dp[j - nums[i]]
```

PERMUTATION COUNT (order matters):

```
dp[0] = 1
for j from 1 to target:                // OUTER = target value
    for each num in nums:              // INNER = try every num, so orderings differ
        if j >= num: dp[j] += dp[j - num]
```

// — 0/1 KNAPSACK — each item at most once

```
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = target; j >= nums[i]; j--)    // DESCENDING: sees OLD dp (before item i)
        dp[j] += dp[j - nums[i]];
}
```

// — UNBOUNDED KNAPSACK — each item reusable

```
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = nums[i]; j <= target; j++)    // ASCENDING: sees NEW dp (after item i already used)
        dp[j] += dp[j - nums[i]];
}
```

// — PERMUTATION COUNT — order matters (Combination Sum IV)

```
int[] dp = new int[target + 1];
dp[0] = 1;
for (int j = 1; j <= target; j++) {          // OUTER: target
    for (int num : nums) {                  // INNER: try all nums (not fixing order)
        if (j >= num) dp[j] += dp[j - num];
    }
}
```

Why descending for 0/1:

`dp[j - nums[i]]` is read **before** `dp[j]` is updated ($j > j - \text{nums}[i]$). So it always sees the dp state from before item `i` was considered — no item is used twice.

Why ascending for unbounded:

`dp[j - nums[i]]` was already updated in this same pass ($j - \text{nums}[i] < j$, processed earlier). This allows item `i` to be picked multiple times.

2D anchor: both loops are `dp[i][j] = dp[i-1][j] + dp[i-?][j-w]` collapsed to 1D — descending keeps `dp[j-w]` on the *previous* row (item `i` unused), ascending pulls it from the *current* row (item `i` reused).

Mnemonic: DESCENDING = **D**istinct (each item once), ASCENDING = **A**gain (reusable).

Flavor = seed + operator (same skeleton):

- count → `dp[0]=1, dp[j] += dp[j-w]`

- feasibility → `dp[0]=true`, `dp[j] = dp[j] || dp[j-w]`
- max-value → `dp[0]=0`, `dp[j] = Math.max(dp[j], dp[j-w] + v)`

Part 3 — 2D Sequence DP

Variables: `dp[i][j]` = the answer for the prefixes `s1[0..i)` (first `i` chars) and `s2[0..j)` (first `j` chars); index `0` = empty prefix. **Pseudocode:**

```
dp = (m+1) x (n+1) table
initialize dp[0][*] and dp[*][0] from the empty-prefix base cases
for i from 1 to m:                // each char of s1
    for j from 1 to n:            // each char of s2
        if s1[i-1] == s2[j-1]:    // current chars match
            dp[i][j] = dp[i-1][j-1] + 1    // extend the diagonal (both prefixes shrink by 1)
        else:
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) // drop a char from one/both side
return dp[m][n]
```

```
// i indexes string/array 1 (1-indexed), j indexes string/array 2
int[][] dp = new int[m + 1][n + 1];
// Init dp[0][*] and dp[*][0] based on problem
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (s1.charAt(i-1) == s2.charAt(j-1)) {
            dp[i][j] = dp[i-1][j-1] + 1;    // characters match
        } else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]);
        }
    }
}
```

Part 4 — Interval DP

Rule: `i` goes right-to-left (ensures shorter/inner intervals are computed first). `j` goes left-to-right from `i+1`. When computing `dp[i][j]`, all `dp[i'][j']` with `i' > i` or `j' < j` are already done.

Variables: `dp[i][j]` = the answer for the interval `[i, j]` (left endpoint `i`, right endpoint `j`). **Pseudocode:**

```
dp = n x n table
for each i: dp[i][i] = base_case    // length-1 intervals
for i from n-1 down to 0:          // i RIGHT-TO-LEFT so dp[i+1][*] (inner) is ready
    for j from i+1 to n-1:          // j LEFT-TO-RIGHT from i+1 so dp[i][j-1] is ready
        // safe to read dp[i+1][j], dp[i][j-1], dp[i+1][j-1] (all strictly smaller intervals)
        dp[i][j] = ...transition over interval [i,j]...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case;    // single elements
for (int i = n - 1; i >= 0; i--) {                  // right-to-left
    for (int j = i + 1; j < n; j++) {                // left-to-right from i+1
        // dp[i][j] can safely use dp[i+1][j], dp[i][j-1], dp[i+1][j-1]
        dp[i][j] = ...;
    }
}
```

Alternative (forward `i`, inner `j` descending): `i` is the right endpoint going left-to-right; `j` is the left endpoint sweeping back from `i-1`. Inner intervals (larger `j`, smaller `i`) are already filled.

Variables: `dp[i][j]` = the answer for the interval `[j, i]` (right endpoint `i`, left endpoint `j`). **Pseudocode:**

```
dp = n x n table
for each i: dp[i][i] = base_case           // length-1 intervals
for i from 0 to n-1:                       // i = RIGHT endpoint, FORWARD left-to-right
    for j from i-1 down to 0:              // j = LEFT endpoint, DESCENDING from i-1 to 0
        // safe to read dp[i-1][j], dp[i][j+1], dp[i-1][j+1] (all shorter intervals already filled)
        dp[i][j] = ...transition over interval [j,i]...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case; // single elements
for (int i = 0; i < n; i++) {
    for (int j = i - 1; j >= 0; j--) {           // j = left endpoint, from i-1 down to 0
        // dp[i][j] can safely use dp[i-1][j], dp[i][j+1], dp[i-1][j+1]
        dp[i][j] = ...;
    }
}
```

Alternative (triangular fill, column base case): used when `dp[i][j]` depends only on the previous row/column (e.g. counting problems). Fill row by row with `j` from `0` to `i`.

Variables: `dp[i][j]` = the count/answer for the `(i, j)` subproblem, defined only for `j <= i` (lower-triangular).

Pseudocode:

```
dp = n x n table
for each i: dp[i][0] = 1                   // column base case; dp[0][j>0] = 0
for i from 1 to n-1:                       // fill ROW BY ROW, top to bottom
    for j from 1 to i:                     // j from 1 up to i (triangular, j <= i)
        // safe to read dp[i-1][j], dp[i][j-1], dp[i-1][j-1] (prior row / earlier in row)
        dp[i][j] = ...transition...
```

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][0] = 1; // base: dp[i][0] = 1; dp[0][j] = 0 for j > 0
for (int i = 1; i < n; i++) {
    for (int j = 1; j <= i; j++) {         // j from 1 to i
        // dp[i][j] can safely use dp[i-1][j], dp[i][j-1], dp[i-1][j-1]
        dp[i][j] = ...;
    }
}
```

Part 5 — Grid DP

Variables: `dp[i][j]` = best (path count / cost) to reach cell `(i, j)` moving only right/down; `memo[i][j]` = longest path starting from `(i, j)` for the all-direction DFS variant; `dr / dc` = 4-direction deltas. **Pseudocode:**

```

dr = {1,-1,0,0}; dc = {0,0,1,-1}           // DOWN UP RIGHT LEFT

// Standard grid (only right/down – DAG, no cycle):
dp = rows x cols table
initialize first row and first column
for i from 0 to rows-1:
    for j from 0 to cols-1:
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1])    // combine top & left

// All-direction grid (memoized DFS for increasing/decreasing paths):
memo = rows x cols table
for i from 0 to rows-1:
    for j from 0 to cols-1:
        dfs(grid, i, j, memo, dr, dc)           // each cell caches its longest path

```

```

int[] dr = {1,-1,0,0}, dc = {0,0,1,-1};    // DOWN UP RIGHT LEFT

// Standard grid (only move right/down – DAG, no cycle):
int[][] dp = new int[rows][cols];
// initialize first row and first column
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]);

// All-direction grid (memoized DFS for increasing/decreasing paths):
int[][] memo = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dfs(grid, i, j, memo, dr, dc);

```

Part 6 — State Machine DP (Stock Problems)

```

cash      = max profit when NOT holding (free to buy or idle)
hold      = max profit when HOLDING stock (bought it at some cost)
cooldown  = max profit right after selling (can't buy today)

```

Part 6 — State Machine DP (Stock Problems)

	buy	sell	re-buy condition
#121 (1 tx):	hold = max(hold, -price)		no re-buy (ignore cash)
#122 (unlimited):	hold = max(hold, cash - price)		re-buy with full cash
#123 (2 tx):	chain: buy2 uses sell1 cash		4 states chained
#309 (cooldown):	hold = max(hold, cooldown - price)		must wait 1 day
#714 (fee):	hold = max(hold, cash - price)		pay fee on sell

Bit Manipulation `bit_manipulation.md`

Canonical Template

Variables: `result` = XOR accumulator (lone survivor) · `n` = number being inspected/edited · `i` = bit position · `count` = set-bit counter · `mask` = 26-bit letter-presence set · masks: `1 << i` selects bit `i`, `n & (n-1)` clears lowest set bit, `n &`

(-n) isolates lowest set bit **Pseudocode:**

XOR CANCEL:

```
result = 0
for each num in nums: result = result XOR num    (pairs cancel, lone survives)
```

BIT OPERATIONS on n at position i:

```
isSet  = bit i of n is 1      -> (n >> i) & 1 == 1
set    = turn bit i on       -> n | (1 << i)
clear  = turn bit i off      -> n & ~(1 << i)
toggle = flip bit i          -> n ^ (1 << i)
lowest = isolate lowest set bit -> n & (-n)
isPow2 = n > 0 and clearing lowest set bit gives 0
```

COUNT SET BITS (Brian Kernighan):

```
count = 0
while n != 0: count = count + 1; n = n & (n-1)    (each step removes one set bit)
```

BITMASK AS SET:

```
mask = 0
for each char c in word: mask = mask | (1 << (c - 'a'))
two words share no letter when (mask[i] & mask[j]) == 0
```

```
// bits are a set/counter - XOR cancels duplicates, AND/OR/shift edit individual bits. - WHEN: "appear
// — XOR CANCEL — pairs cancel; lone element survives
int result = 0;
for (int num : nums) result ^= num;

// — BIT OPERATIONS —
boolean isSet = ((n >> i) & 1) == 1;    // check if bit i is set
int set      = n | (1 << i);           // set bit i
int clear    = n & ~(1 << i);          // clear bit i
int toggle   = n ^ (1 << i);           // toggle bit i
int lowest   = n & (-n);               // isolate lowest set bit
boolean isPow2 = n > 0 && (n & (n-1)) == 0;

// — COUNT SET BITS — Brian Kernighan: each iteration removes one set bit
// WHY n & (n-1) clears the lowest set bit:
//   n      = ...1000 (lowest set bit at this position)
//   n-1    = ...0111 (borrow flips that bit to 0 and all lower bits to 1)
//   n&(n-1)=...0000 (AND keeps higher bits, wipes the lowest set bit + below)
int count = 0;
while (n != 0) { count++; n &= (n - 1); } // n & (n-1) clears lowest set bit

// — BITMASK AS SET — 26-bit integer represents presence of 26 letters
int mask = 0;
for (char c : word.toCharArray()) mask |= (1 << (c - 'a'));
// check no overlap between two words:
if ((mask[i] & mask[j]) == 0) { /* no shared letter */ }
```

Variations

Variables: ones / twos = bits seen 1 / 2 times mod 3 · xor = $a \oplus b$ of the two uniques · diff = lowest differing bit
(xor & -xor) · a / b = the two unique values · dp[] = set-bit counts by number · shift = bits dropped to reach
common prefix · carry = AND-derived carry bits **Pseudocode:**

VARIATION 1 (mod-3 state machine, Single Number II):

```
ones = 0; twos = 0
for each num: ones = (ones XOR num) AND NOT twos; twos = (twos XOR num) AND NOT ones
after loop, ones holds the element appearing once
```

VARIATION 2 (split XOR into two groups, Single Number III):

```
xor = XOR of all nums                (= a XOR b)
diff = xor AND (-xor)                (lowest bit where a and b differ)
a = 0; for each num: if (num AND diff) != 0: a = a XOR num    (XOR one group)
b = xor XOR a                        (the other unique)
```

VARIATION 3 (DP relation, Counting Bits):

```
dp = array of size n+1
for i = 1..n: dp[i] = dp[i >> 1] + (i AND 1)    (drop last bit, add it back)
```

VARIATION 4 (common prefix, Bitwise AND of Range):

```
shift = 0
while left != right: left >>= 1; right >>= 1; shift = shift + 1
return left << shift                (shared high-bit prefix restored)
```

VARIATION 5 (carry loop, Sum of Two Integers):

```
while b != 0:
    carry = a AND b                (positions that carry)
    a = a XOR b                    (sum without carry)
    b = carry << 1                (carry moves one bit left)
return a
```

```

// VARIATION 1: mod-3 state machine (Single Number II)
// MENTAL MODEL: extend XOR from mod-2 (cancel pairs) to mod-3 (cancel triples);
//                ones/twos track bits seen 1 or 2 times mod 3.
// Instead of XOR canceling pairs (mod 2), cancel triples (mod 3)
// ones = bits seen exactly 1 mod 3 times; twos = bits seen exactly 2 mod 3 times
int ones = 0, twos = 0;
for (int num : nums) {
    ones = (ones ^ num) & ~twos;    // ← add to ones if not already in twos
    twos = (twos ^ num) & ~ones;    // ← add to twos if not already in ones
}
// After all: ones holds the element appearing once (0 mod 3 remainder = survive)

// VARIATION 2: split XOR into two groups (Single Number III)
// Two unique elements a, b. XOR of all = a ^ b.
// Use lowest set bit of (a^b) to split nums into two groups → each group has one unique.
int xor = 0;
for (int num : nums) xor ^= num;    // xor = a ^ b
int diff = xor & (-xor);             // ← lowest set bit that differs between a and b
int a = 0;
for (int num : nums)
    if ((num & diff) != 0) a ^= num; // ← XOR one group → one unique element
int b = xor ^ a;                     // ← the other unique element

// VARIATION 3: DP relation (Counting Bits)
// dp[i] = dp[i/2] + last bit of i
// i >> 1 = i without last bit (same or fewer ones); last bit = i & 1
int[] dp = new int[n + 1];
for (int i = 1; i <= n; i++)
    dp[i] = dp[i >> 1] + (i & 1);    // ← right-shift halves the problem

// VARIATION 4: common prefix (Bitwise AND of Range)
// AND of any range [left, right]: bits that differ between left and right get cleared.
// Right-shift both until equal → find shared high-bit prefix → shift back.
int shift = 0;
while (left != right) { left >>= 1; right >>= 1; shift++; }
return left << shift;               // ← shift back to original magnitude

// VARIATION 5: carry loop (Sum of Two Integers)
// XOR gives bit sum without carry; AND shifted left gives carry.
// Repeat until no carry remains.
int add(int a, int b) {
    while (b != 0) {
        int carry = a & b;           // ← carry bits
        a = a ^ b;                   // ← partial sum (no carry)
        b = carry << 1;              // ← carry propagated one position left
    }
    return a;
}

```

Design `design.md`

Canonical Template — HashMap + Doubly Linked List (LRU)

Variables: `map` = key -> CacheNode · `head` / `tail` = sentinels (head side = MRU, tail side = LRU) ·

`node.previous` / `node.next` = doubly linked neighbors **Pseudocode:**

```

CacheNode: key, value, previous, next
init: head <=> tail (head.next = tail; tail.previous = head)

remove(node):
    node.previous.next = node.next
    node.next.previous = node.previous

insertAfterHead(node):           # mark as most-recently-used
    link node between head and head.next

evict LRU:
    lru = tail.previous           # node just before tail
    remove(lru); map.remove(lru.key)

```

```

// HashMap gives O(1) lookup; the doubly linked list orders by recency so the LRU is always one hop fro
// CacheNode for doubly linked list (named CacheNode to distinguish from ListNode contexts)
class CacheNode {
    int key, value;
    CacheNode previous, next;
    CacheNode(int key, int value) { this.key = key; this.value = value; }
}

// — SENTINEL NODES — head (MRU side) ↔ ... ↔ tail (LRU side)
CacheNode head = new CacheNode(0, 0), tail = new CacheNode(0, 0);
// Initialize: head <=> tail
head.next = tail; tail.previous = head;

// — REMOVE NODE —
void remove(CacheNode node) {
    node.previous.next = node.next;
    node.next.previous = node.previous;
}

// — INSERT AFTER HEAD (= mark as MRU) —
void insertAfterHead(CacheNode node) {
    node.next = head.next;
    node.previous = head;
    head.next.previous = node;
    head.next = node;
}

// — LRU EVICTION — tail.previous is LRU
CacheNode lru = tail.previous;
remove(lru);
map.remove(lru.key);

```

Variations

Variables: `keyToVal` / `keyToFreq` = key lookups · `freqToKeys` = freq -> ordered keys at that freq · `minFreq` = smallest live frequency · (RandomizedSet) `list` = values array · `map` = val -> index **Pseudocode:**

VARIATION 1 (LFU): maps key->value, key->freq, freq->LinkedHashSet<key>, plus minFreq
on access: freq++; move key from freqToKeys[old] to freqToKeys[new]
on evict: remove first key (oldest) from freqToKeys[minFreq]

VARIATION 2 (RandomizedSet): ArrayList of values + HashMap val->index
remove: swap target with last element, update map, drop last
getRandom: list[random index]

```
// VARIATION 1: LFU – adds a frequency dimension
// Need: key->value, key->freq, freq->orderedSet<key>, and minFreq
// When a key is accessed: freq++, move from freqToKeys[oldFreq] to freqToKeys[newFreq]
// On eviction: remove first element from freqToKeys[minFreq]
// LinkedHashSet preserves insertion order → oldest at front for same frequency

// VARIATION 2: Insert Delete GetRandom – use array for O(1) random
// ArrayList stores values; HashMap stores val->index in array
// On remove: swap target with last element, update map, remove last
// On getRandom: return list.get(random.nextInt(list.size()))
```

Math `math.md`

Canonical Templates

Variables: `n` = the number being processed · `digit` = last digit peeled · `x` = base in fast power · `result` = running product/accumulator · `b` = the base for conversion · `composite[]` = sieve marks **Pseudocode:**

DIGIT EXTRACTION:

```
while n is not zero:
    digit = remainder of n divided by 10
    drop the last digit of n
    use digit
```

FAST POWER (x^n):

```
result starts at 1
while exponent n still has bits:
    if the lowest bit is set, fold the current x into result
    square x to reach the next power
    shift n right to consume that bit
return result
```

BASE CONVERSION (number $\rightarrow$ digits):

```
while num is positive:
    append num mod b
    divide num by b
reverse the collected digits
```

BASE CONVERSION (digits $\rightarrow$ number):

```
value starts at 0
for each digit: value = value * b + digit
```

SIEVE:

```
make a composite[] flag array up to n
for i from 2 while  $i*i < n$ :
    if i already marked composite, skip it
    mark every multiple of i starting at  $i*i$  as composite
```



```

// MENTAL MODEL: number = sequence of digits in some base; iterate by repeatedly
//                peeling the last digit (% base) and shifting (/ base).
// WHEN: "reverse digits", "x^n", "base conversion", "count primes", "nth divisible-by".

// — DIGIT EXTRACTION LOOP — peel digits right-to-left
while (n != 0) {
    int digit = n % 10;    // last digit
    n /= 10;               // drop last digit
    // ... use digit ...
}

// — FAST POWER (exponentiation by squaring) —
// WHY: x^n needs only O(log n) multiplications by squaring the base
//      and consuming one exponent bit per step.
// x^13 = x^8 * x^4 * x^1  (13 = 1101 in binary)
double fastPow(double x, long n) {
    double result = 1;
    while (n > 0) {
        if ((n & 1) == 1) result *= x;    // bit set → include this power of x
        x *= x;                          // square the base for the next bit
        n >>= 1;                          // consume one exponent bit
    }
    return result;
}

// — BASE CONVERSION — generic base b, 0-indexed
// number → digits: peel with % b, shift with / b, collect, then reverse
StringBuilder builder = new StringBuilder();
while (num > 0) {
    builder.append(num % b);
    num /= b;
}
builder.reverse();
// digits → number: Horner's rule
int value = 0;
for (int digit : digits) value = value * b + digit;

// — SIEVE OF ERATOSTHENES — mark composites up to n
// WHY start at i*i: any multiple k*i with k < i was already marked by k.
boolean[] composite = new boolean[n];
for (int i = 2; i * i < n; i++) {
    if (composite[i]) continue;
    for (int j = i * i; j < n; j += i)    // start at i*i, step by i
        composite[j] = true;
}

```